



EXAM PAPERS PRACTICE

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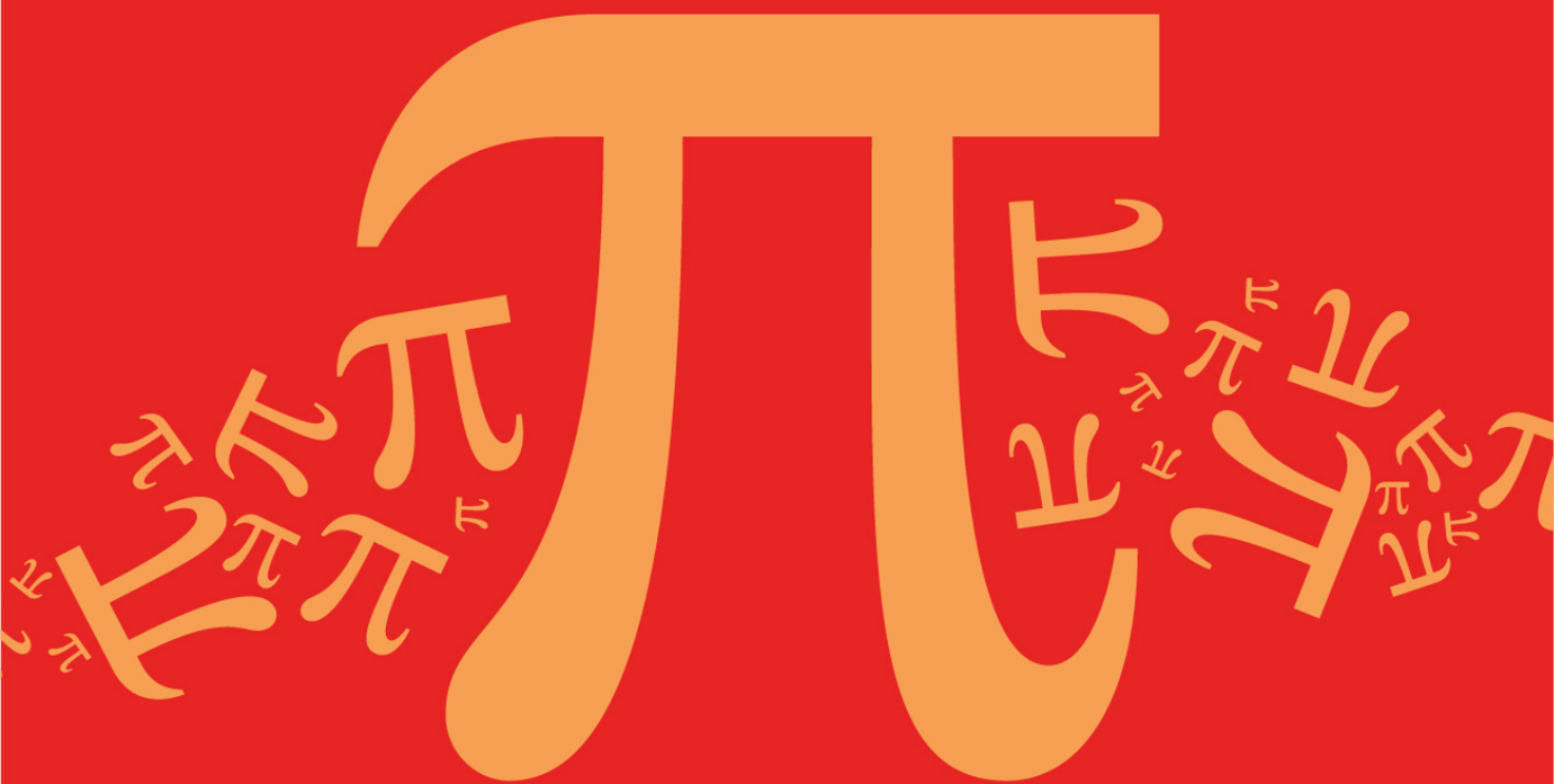
Detailed mark scheme

Suitable for all boards

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thoroughly prepare you

AQA A Level Further Mathematics

(7367)



Topic

C: Matrices

Sub topic

C1: Manipulate Matrices

Mark Scheme

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Mark Scheme

Topic

C: Matrices

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C1: Manipulate Matrices

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1a	B1	AC
Total 1 mark			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses factorisation or pre-multiplication to isolate A	AO3.1a	M1	$\mathbf{AB} - 2\mathbf{A} = \mathbf{I}$ $\mathbf{A}(\mathbf{B} - 2\mathbf{I}) = \mathbf{I}$
Deduces A in terms of B and I . Could be implied by sight of $\begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}$ with attempt to invert.	AO2.2a	A1	$\mathbf{A} = (\mathbf{B} - 2\mathbf{I})^{-1}$
Obtains correct matrix A .	AO1.1b	A1	$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}^{-1}$ $\mathbf{A} = \frac{1}{-2} \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$

ALT

Marking Instructions	AO	Marks	Typical Solution
Sets up four equations with at least three correct.	AO3.1a	M1	Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2 \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $3a - 4b = 1 + 2a$ and $3c - 4d = 0 + 2c$ and $-2a + 8b = 0 + 2b$ and $-2c + 8d = 1 + 2d$
Deduces at least two correct elements of A . Note: Two correct elements from just two correct equations can score M1A1.	AO2.2a	A1	$a = 4b + 1$ and $c = 4d$ $6b = 2a$ and $6d = 2c + 1$ $\therefore 3b = 4b + 1$ and $6d = 2(4d) + 1$ $-1 = b$ and $-1 = 2d \Rightarrow d = -\frac{1}{2}$ $\therefore a = 4 \times -1 + 1$ and $c = 4 \times -\frac{1}{2}$

			$a = -3$ and $c = -2$
Obtains correct matrix A	AO1.1b	A1	$\therefore \mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$
Total 3 marks			

Q3.

Marking Instructions	AO	Marks	Typical Solution
Uses proof by induction and investigates formula for $n = 1$ and $n = k$ (must see evidence of both $n = 1$ and $n = k$ being considered)	AO3.1a	M1	Using induction method, Let $P(n)$ be the statement $\mathbf{M}^n = \begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$
Demonstrates that formula is true for $n = 1$	AO1.1b	A1	For $n = 1$ $\begin{bmatrix} 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \mathbf{M}^1$
States assumption that formula true for $n = k$ and uses $\mathbf{M}^{k+1} = \mathbf{M} \times \mathbf{M}^k$	AO2.1	R1	$\therefore P(1)$ is true Assume $P(k)$ is true $\mathbf{M}^{k+1} = \mathbf{M} \times \mathbf{M}^k$
Deduces that formula is also true for $n = k + 1$ from correct working	AO2.2a	R1	$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{bmatrix}$ (since $P(k)$ is true)
Completes a rigorous argument and explains how their argument proves the required result. AG	AO2.4	R1	$= \begin{bmatrix} (3^{k-1} + 3^{k-1} + 3^{k-1}) & (3^{k-1} + \dots) & (3^{k-1} + \dots) \\ (3^{k-1} + 3^{k-1} + 3^{k-1}) & (3^{k-1} + \dots) & (3^{k-1} + \dots) \\ (3^{k-1} + 3^{k-1} + 3^{k-1}) & (3^{k-1} + \dots) & (3^{k-1} + \dots) \end{bmatrix}$ But $3^{k-1} + 3^{k-1} + 3^{k-1} = 3 \times 3^{k-1} = 3^k$ $\mathbf{M}^{k+1} = \begin{bmatrix} 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \end{bmatrix}$ $\therefore \mathbf{M}^{k+1} = \begin{bmatrix} 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \end{bmatrix}$ $\therefore P(k + 1)$ is true Since $P(1)$ is true

			and $P(k) \Rightarrow P(k + 1)$ hence, by induction, $P(n)$ is true for all $n \in \mathbb{N}$ AG
Total 5 marks			

Q4.

(a) (i) $X^2 = \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}; \quad (m \Rightarrow) 7$
(m $\Rightarrow$) 7 or 7 as top left element of X^2

B1

1

(ii) $X^3 = \begin{bmatrix} 13 & 14 \\ 21 & 6 \end{bmatrix};$
At least 2 elements correct

M1

$7X = \begin{bmatrix} 7 & 14 \\ 21 & 0 \end{bmatrix}$
PI

B1

$X^3 - 7X = \begin{bmatrix} 13-7 & 14-14 \\ 21-21 & 6-0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$

Ft on c's m value

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A1F

$= 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 6I$

CSO Accept either form but at least one must be shown explicitly

A1

4

(b) (i) Reflection in the x-axis
OE

B1

1

(ii) $B = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

Either OE. For M mark, accept dec. equiv.

(at least 3sf) for $\frac{1}{\sqrt{2}}$

M1

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

NMS SC1 for $k = \frac{1}{\sqrt{2}}$ or better.

A1

2

(iii) $\mathbf{AB} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = k \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Attempt to find $\mathbf{AB} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

M1

$$= k \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \left\{ \text{or } k \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$$

Either $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

A1

$$= k \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

Completing the matrix mult. to reach a 2×1 matrix

m1

(Image of P is the point) $\left(-\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$

CSO

SC Wrong order, works with $\mathbf{BA} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$,
mark out of a max of M1A0 m1A0

A1

4

[12]

Q5.

(a) $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

$$\begin{bmatrix} -1 & -2 & -4 \\ 5 & 6 & 4 \\ 10 & 10 & 9 \end{bmatrix}$$

$$p = -1$$

p-value

B1

$$q = 10$$

q-value

B1

2

(b) $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = 0$ multiply by $\mathbf{A}^{-1}$

$$(\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I})\mathbf{A}^{-1} = (0)\mathbf{A}^{-1}$$

Multiplication by $\mathbf{A}^{-1}$

M1

$$\mathbf{A}^3\mathbf{A}^{-1} - 6\mathbf{A}^2\mathbf{A}^{-1} + 11\mathbf{A}\mathbf{A}^{-1} - 6\mathbf{I}\mathbf{A}^{-1} = 0$$

$$\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I} - 6\mathbf{A}^{-1} = 0$$

$$6\mathbf{A}^{-1} = \mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I}$$

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I})$$

AG

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A1

2

(c) $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix}$

$$r = 4$$

r-value

B1

$$s = -2$$

s-value

B1

2

(d) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix} \begin{bmatrix} k \\ 5 \\ 7 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k - 10 + 14 \\ -k + 25 - 14 \\ -2k - 10 + 14 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k + 4 \\ 11 - k \\ 4 - 2k \end{bmatrix}$

use of $\mathbf{A}^{-1} \mathbf{v}$ – one row correct

M1

$$x = \frac{2k+2}{3}, y = \frac{11-k}{6}, z = \frac{2-k}{3}$$

correct solution for one variable

A1

all correct CAO

A1

Alternative:

If solving equations by elimination, M1 A1 for correct solution for one variable, A1 all correct

3

[9]

Q6.

(a) (i) $\mathbf{A} - \mathbf{B} = \begin{bmatrix} p-3 & 1 \\ 2 & p-3 \end{bmatrix}$

B1

1

(ii) $\mathbf{AB} = \begin{bmatrix} p & 2 \\ 4 & p \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3p+4 & p+6 \\ 12+2p & 4+3p \end{bmatrix}$

Finding $\mathbf{AB}$ and at least 2 elements correct

M1

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A1

2

(b) $\mathbf{A} - \mathbf{B} + \mathbf{AB} = \begin{bmatrix} 4p+1 & p+7 \\ 14+2p & 1+4p \end{bmatrix}$

Only ft if all matrices are 2 by 2

PI by later correct work

B1F

$$\mathbf{A} - \mathbf{B} + \mathbf{AB} = k \mathbf{I} = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathbf{I}$ used as or equated to $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ at some stage

B1

$$(p + 7 = 0, 14 + 2p = 0 \Rightarrow) p = -7$$

$p = -7$ provided it gives the relevant two zero elements

B1

$$p = -7 \Rightarrow \mathbf{A} - \mathbf{B} + \mathbf{AB} = \begin{bmatrix} -27 & 0 \\ 0 & -27 \end{bmatrix} = -27 \mathbf{I}$$

$$\Rightarrow k = -27$$

CSO

Either $-27 \mathbf{I}$ (no earlier errors) for B1

OR $k = -27$ with either $\begin{bmatrix} -27 & 0 \\ 0 & -27 \end{bmatrix}$ or $27 \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ or

$-27 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ seen before (no earlier errors) for B1

B1

4

[7]

Q7.

(a) $\text{Det}(\mathbf{AB}) = (4)(1) - (1)(1) = 3 \neq 0$

Must state non-zero or $\neq 0$

B1

1

(b) Matrix of cofactors $\begin{bmatrix} 0 & 0 & 3 \\ 4 & -1 & 8-4k \\ -1 & 1 & k-8 \end{bmatrix}$

Attempt at matrix of cofactors – at least five correct entries

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M1

Fully correct matrix of cofactors

A1

$$(\mathbf{AB})^{-1} = \frac{1}{3} \begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix}$$

Their cofactor matrix transposed correctly

M1

At least five correct = A1 (exclude effect of determinant)

All entries fully correct = A2

A2,1

5

(c) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

$$\Rightarrow \mathbf{B}^{-1} = (\mathbf{AB})^{-1} \mathbf{A}$$

$$B^{-1} = \frac{1}{3} \begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix} \begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$$

Use of $(AB)^{-1}$ and A multiplied

M1

Correct order of multiplication

A1

$$= \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ -3 & 3 & 6 \\ 24 & -6 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 2 \\ 8 & -2 & -8 \end{bmatrix}$$

All correct = A2

5+ correct = A1 (exclude effect of determinant)

NB – if an attempt is made to find B by setting up a system of simultaneous equations then

M1 – 9 correct equations used

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & \frac{1}{2} \\ \frac{3}{2} & \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

A1 for $B =$

Final A2 as above

A2,1

4

[10]

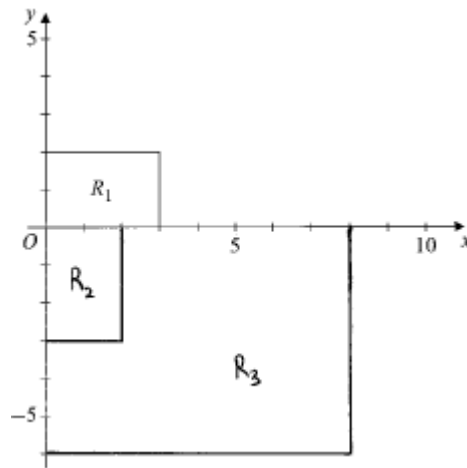
Q8.

(a) (i)

Rectangle with vertices $(0, 0)$, $(0, -3)$, $(2, -3)$, $(2, 0)$

B1

1



(ii)

Rectangle with vertices **either** whose x -coords are c 's (a)(i) x -coord vertices multiplied by 4 or whose y -coords are c 's (a)(i) y -coord vertices multiplied by 2

M1

A2 if rectangle with vertices $(0, 0), (0, -6), (8, -6), (8, 0)$ (A1 if either the four x -coords are correct or the four y -coords are correct)

A2,1

3

(b) (i) Matrix is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

B1

1

(ii) $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} =$

Attempt to multiply $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ with c 's (b)(i) matrix in **either** order.

M1

Multiplication in **correct** order with at least two of the four ft multiplications carried out correctly.

m1

$$\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$$

For $\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$

NMS $\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$ scores B3

$\begin{bmatrix} 0 & 2 \\ -4 & 0 \end{bmatrix}$ scores B1

A1

3

[8]

Q9.

(a)

$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

If A1 not scored, award M1A0 for all correct entries expressed in trig form eg

M1

$$\begin{bmatrix} \cos 135 & -\sin 135 \\ \sin 135 & \cos 135 \end{bmatrix}$$

A1

2

(b) (i)

M =

$$\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \sqrt{2} \times \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \left(\begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \right)$$

Or better

PI by cand. having both a correct scale factor of enlargement and a correct corresponding angle of rotation

M1

Scale factor of enlargement is $\sqrt{2}$

SF = $\sqrt{2}$ OE surd form

A1

Angle of rotation is 135 (degrees anticlockwise)

Angle = 135 OE eg -225

If M0 give B1 for SF = $\sqrt{2}$ OE surd and

B1 for angle = 135 OE

A1

3

(ii) For M^2 , SF of enlargement = 2

OE If incorrect, ft on [c's SF in (b)(i)]²

OE, eg -90(degrees), eg 90 (degrees) clockwise

B1F

Angle of rotation is 270 (degrees anticlockwise)

If incorrect, ft on $2 \times c$'s angle in (b)(i)

(neither B1F B1 nor B1 B1F is possible)

B1F

2

$$(iii) \quad \mathbf{M}^2 = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

For complete method (matrix calculation or geometrical reasoning)

$$\mathbf{M}^4 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}$$

Matrix for $\mathbf{M}^2$ could be seen earlier

(M0 if >1 independent error in matrix multiplication)

Geometrically SF = 4, rotation angle = 540

OE scores M1 and completion scores A1

M1

$$\mathbf{M}^4 - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -4\mathbf{I}$$

Either of these two forms convincingly shown

A1

2

$$(iv) \quad \mathbf{M}^{2012} = (\mathbf{M}^4)^{503} = (-4\mathbf{I})^{503} = -(2^2)^{503}\mathbf{I} = -2^{1006}\mathbf{I}$$

OE Fully explained, algebraically from $(-4\mathbf{I})^{503}$, or geometrically

E1

$$\mathbf{M}^{2012} = -2^{1006}\mathbf{I}$$

$$\mathbf{M}^{2012} = -2^{1006}\mathbf{I} (n = 1006) \quad (B0 \text{ if FIW})$$

B1

Geometrically: $\mathbf{M}^{2012}$ represents an enlargement with SF 2^{1006} followed by a rotation of angle $2012 \times 135^\circ$ ie 754.5 revolutions, being equivalent to rotation of 180° ie matrix is $-\mathbf{I}$ so

$$\mathbf{M}^{2012} = -2^{1006}\mathbf{I}$$

2

[11]

Q10.

$$(a) \quad (i) \quad \mathbf{A}^T = \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$$

B1

Good multiplication attempt at $\mathbf{A} \mathbf{A}^T$

If they multiply $\mathbf{A}^T \mathbf{A}$ instead, they can score B1 M1 A0 A0 A1 A1

A1A1

M1

$$= \begin{bmatrix} k^2 + 5 & 3k + 2 & 3k + 2 \\ 3k + 2 & k^2 + 5 & 3k + 2 \\ 3k + 2 & 3k + 2 & k^2 + 5 \end{bmatrix}$$

Main diagonal correct

A1

All others correct

A1

$$k = -\frac{2}{3} \quad m = 5\frac{4}{9} \text{ or } \frac{49}{9}$$

6

(ii)
$$\begin{bmatrix} -\frac{2}{3} & 1 & 2 \\ 2 & -\frac{2}{3} & 1 \\ 1 & 2 & -\frac{2}{3} \end{bmatrix}^{-1} = \frac{9}{49} \begin{bmatrix} -\frac{2}{3} & 2 & 1 \\ 1 & -\frac{2}{3} & 2 \\ 2 & 1 & -\frac{2}{3} \end{bmatrix}$$

ft their m and their k

B1

Accept $\frac{9}{49} \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$ since the value of k is now known,
but not just $\frac{9}{49} \mathbf{A}^T$

Decimal version (correct to at least 3sf) is also ok:

$$\begin{bmatrix} -0.122 & 0.367 & 0.184 \\ 0.184 & -0.122 & 0.367 \\ 0.367 & 0.184 & -0.122 \end{bmatrix}$$

1

(b) (i) $\det \mathbf{A} = k^3 - 6k + 9$
Good attempt (cubic)

M1

A1

2

(ii) $\det \mathbf{A} = (k + 3)(k^2 - 3k + 3)$
Factorisation attempt

M1

$$k = -3$$

A1

$$\Delta = 9 - 12$$

B1

Special Cases

$k = -3$ but no working

(B1)

For all 3 roots, -3 and $\frac{3+i\sqrt{3}}{2}$ given
with no supporting working

(B3)

3

(iii) Replacing k by $(k - 7)$

NOT just in the determinant form (ie starting again)

M1

$k = 4$ (however obtained)
ft their previous $k + 7$

A1

2

[14]

Q11.

(a) (i) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

B1

1

(ii) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

B1

1

(b) (i) $\mathbf{AB} = \begin{bmatrix} -20 & 14 \\ 14 & -10 \end{bmatrix}$

M1A0 if 3 entries correct

M1A1

2

(ii) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$

B1

$$(\mathbf{A} + \mathbf{B})^2 = \begin{bmatrix} -25 & 0 \\ 0 & -25 \end{bmatrix}$$

B1

$$\dots = -25\mathbf{I}$$

ft if c's $(\mathbf{A} + \mathbf{B})^2$ is of the form $k\mathbf{I}$

B1F

3

- (c) (i) Rot'n 90° clockwise, enlargem't SF 5
OE

B2, 1

2

- (ii) Rotation 180° , enlargement SF 25
*Accept 'enlargement SF -25';
ft wrong value of k*

B2, 1F

2

- (iii) Enlargement SF 625
B1 for pure enlargement; ft ditto

B2, 1F

2

EXAM PAPERS PRACTICE [13]

Q12.

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$$\mathbf{X}^2 = \begin{bmatrix} 23 & 1 & -3 \\ 2 & 24 & 3 \\ -4 & 2 & 19 \end{bmatrix}$$

- (a) (i)

≥ 5 correct for the M

M1

All 9 correct for the A

A1

$$\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I} \quad \text{i.e. } k = 20$$

Shown legitimately

A1

3

- (ii) Mult⁹. $\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I}$ by $\mathbf{X}^{-1}$

M1

Re-arranging $\mathbf{X} - \mathbf{I} = 20 \mathbf{X}^{-1}$

to get $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$

Legitimately

A1

2

(b) $\mathbf{X}^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 1 & -3 \\ 2 & 3 & 3 \\ -4 & 2 & -2 \end{bmatrix}$

Noted or used

B1

$(\mathbf{XY})^{-1} = \mathbf{Y}^{-1} \mathbf{X}^{-1}$

Incl. attempt at the multn.

M1

$\begin{bmatrix} 6 & 3 & -9 \\ 2 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$

A1

3

EXAM PAPERS PRACTICE [8]

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Q13.

(a) (i) $\mathbf{A}^2 = \begin{bmatrix} -2 & 2\sqrt{3} \\ -2\sqrt{3} & -2 \end{bmatrix}$

M1 if at least two entries correct

M1A1

2

(ii) $\mathbf{A}^3 = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$

if at least two entries correct

M1

$\dots\dots\dots = 8\mathbf{I}$

A1

2

- (b) (i) $\mathbf{A}^3$ gives enlargement with SF 8 (centre the origin)
M1 for enlargement (only);

ft wrong value for k

M1A1F

2

- (ii) Enlargement and rotation
Some detail needed

M1

Enlargement scale factor 2

A1

Rotation through 120° (antic'wise)

A1

3

[9]

Q14.

(a) $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

stated or used at any stage

B1

Attempt at $(A - I)^2$

with at most one numerical error

M1

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$$(A - I)^2 = \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} = 12I$$

A1

3

(b) $(A - B) = \begin{bmatrix} 0 & 1 \\ 3 - p & 0 \end{bmatrix}$

B1

$$(A - B)^2 = \begin{bmatrix} 3 - p & 0 \\ 0 & 3 - p \end{bmatrix}$$

M1A0 if 3 entries correct

M1A1

$\dots = (A - I)^2$ for $p = -9$

ft wrong value of k

A1F

4

[7]

Q15.

(a)
$$\begin{bmatrix} 32 - 4t & 0 & 0 \\ 98 - 2t^2 & 4t - 24 & 2t - 14 \\ 28 - 4t & 0 & 4 \end{bmatrix}$$

Decent attempt at AB

M1

≥ 5 correct

A1

all correct

A1

$t = 7$

Allow this ft if only 1 or 2 elements of AB incorrect

A1

$AB = 4I$

Must be from AB completely correct

A1

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(b)
$$\mathbf{A}^{-1} = \frac{1}{4} \begin{bmatrix} 115 & -4 & -1 \\ -14 & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

ft 1/det if related to B

B1

CAO; must be "deduced"

$\frac{1}{4} \mathbf{B}$
NB " $\frac{1}{4}$ " scores B1 only

B1

2

[7]

Q16.

(a)
$$\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} x \\ 2x+1 \end{bmatrix} = \begin{bmatrix} x-1 \\ -x+2 \end{bmatrix}$$

Use of $\begin{bmatrix} x \\ 2x+1 \end{bmatrix}$; either term correct

M1A1

giving $y = 1 - x$
CSO

A1

3

(b) Finding $\det(\mathbf{B}) (= 1)$ and mult⁹. by 4.5

M1

= 4.5 cm²

ft

A1

2

(c) $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
AB attempted

M1A1

A1

This is a shear

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parallel to the x-axis

A1

mapping (eg) (1, 1) to (3, 1)

Any point not on Ox and its image

A1

5

[10]

Q17.

(a) Rotation 45° (anticlockwise) (about O)
M1 for 'rotation'

M1A1

2

(b) Reflection in $y = x \tan 22.5^\circ$
M1 for 'reflection'

M1A1
2

- (c) Rotation 90° (anticlockwise) (about O)
M1 for 'rotation' or correct matrix;
ft wrong angle in (a)

M1A1F
2

- (d) Identity transformation
ft wrong mirror line in (b); B1 for $B^2 = I$

B2,1F
2

- (e) $AB = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
allow M1 if two entries correct

Reflection in $y = x$

M1A1
A1
3

[11]

Q18.

- (a) $AB = \begin{bmatrix} 2x+1 & 2x-1 \\ 8 & 4 \end{bmatrix}$
Good attempt (at least one entry in R_1)

All four correct

M1

A1
2

- (b) $B^T A^T = (AB)^T$ Or $\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ x & 3 \end{bmatrix}$

M1

$$= \begin{bmatrix} 2x+1 & 8 \\ 2x-1 & 4 \end{bmatrix}$$

Ft their (a) Or CAO

A1ft

$2x+1 = 4-4x$ Or $2x-1 = 8x-4$
Ft previous answers

M1

$$x = \frac{1}{2}$$

CAO

A1

Checking / noting $x = \frac{1}{2}$ in other eqn.
Visibly

B1

5

[7]

Q19.

(a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 0 & 2k \\ 2k & 0 \end{bmatrix}$

B1

1

(ii) $\mathbf{A}^2 = \begin{bmatrix} 2k^2 & 0 \\ 0 & 2k^2 \end{bmatrix}$

B1 if three entries correct

B2,1

2

(b) $(\mathbf{A} + \mathbf{B})^2 = \begin{bmatrix} 4k^2 & 0 \\ 0 & 4k^2 \end{bmatrix}$

B1 if three entries correct

B2,1

$\mathbf{B}^2 = \mathbf{A}^2$, hence result

B1B1

4

(c) (i) $\mathbf{A}^2$ is an enlargement (centre O)

M1

with SF 2

Condone $2k^2$

A1

2

(ii) Scale factor is now $\sqrt{2}$
Condone $2k$

B1

Mirror line is $y = x \tan 22\frac{1}{2}^\circ$

M1 A1

3

[12]

Q20.

(a) $\det \mathbf{AB} = 110$

B1

Use of $\det \mathbf{AB} = \det \mathbf{A} \det \mathbf{B}$

M1

$\det \mathbf{B} = 11$

ft their $\det \mathbf{AB} / 10$

A1F

3

(b) $\mathbf{C} = (\mathbf{AB})^T = \begin{bmatrix} 9 & 7 \\ 1 & 13 \end{bmatrix}$

M1A1

$\mathbf{D} = [(\mathbf{BA})^T]^T = \mathbf{BA} = \begin{bmatrix} 14 & 2 \\ 1 & 8 \end{bmatrix}$

For reference:

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 5 & 3 \\ -2 & 1 \end{bmatrix}$$

B1

3

[6]

Q21.

(a) (i) Matrix is $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$
 $M1 \text{ for } \begin{bmatrix} \cos 30^\circ & \sin 30^\circ \\ -\sin 30^\circ & \cos 30^\circ \end{bmatrix} (PI)$

M1A1

2

(ii) Matrix is $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$
 $M1$ for $\begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{bmatrix} (PI)$

M1A1

(b) SF 2, line $y = \frac{1}{\sqrt{3}}x$
 OE

B1B1

2

(c) Attempt at **BA** or **AB**

M1

$BA = \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$

m1 if zeros in correct positions

m1A1

Enlargement SF 4

*ft use of **AB** (answer still 4)*

or after $BA = \begin{bmatrix} 0 & k \\ k & 0 \end{bmatrix}$

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B1F

... and reflection in line $y = x$

ft only from $BA = \begin{bmatrix} 0 & k \\ k & 0 \end{bmatrix}$

B1F

5

[11]

Q22.

(a) $\begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix} \begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} k+14 & -1 \\ 22-k & 3 \end{bmatrix}$

PQ a 2×2 matrix

M1

At least one element in C_1 correct

A1

All correct

A1

3

(b) $\text{Det}(\mathbf{PQ}) = 3k + 42 + 22 - k$

Det of a square matrix attempted and equated to zero

M1

$$= 2k + 64 = 0$$

$$k = -32$$

ft in 2×2 case only (linear eqns.)

A1

2

[5]

Q23.

(a) (i) $\mathbf{M}^2 = \begin{bmatrix} 12 & 0 \\ 0 & 12 \end{bmatrix}$

M1 if zeroes appear in the right places

M1A1

$$= 12\mathbf{I}$$

ft provided of right form

A1F

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(ii) $q \cos 60^\circ = \frac{1}{2}q = \sqrt{3} \Rightarrow q = 2\sqrt{3}$

OE SC $q = 2\sqrt{3}$ NMS 1/3

M1A1

Other entries verified

surd for $\sin 60^\circ$ needed

E1

3

(b) (i) $\text{SF} = q = 2\sqrt{3}$

ft wrong value for q

B1F

1

(ii) Equation is $y = x \tan 30^\circ$

B1

1

(c) $M^4 = 144I$

PI; ft wrong value in (a)(i)

B1F

M^4 gives enlargement SF 144

ft if c's $M^4 = kI$

B1F

2

[10]

Q24.

(a) (i) $M^2 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$

M1

$$= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix}$$

A1

$$\begin{aligned}
 M^2 + 2I &= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\
 &= \begin{bmatrix} 6 & -3 & 3 \\ 3 & 0 & 3 \\ 3 & -3 & 6 \end{bmatrix} = 3M
 \end{aligned}$$

ie $k = 3$

A1

3

(ii) Multiplying by M^{-1}

M1

to get $M + 2M^{-1} = 3I$

ft

A1

so that $M^{-1} = \frac{3}{2}I - \frac{1}{2}M$

ie $a = -\frac{1}{2}$ and $b = \frac{3}{2}$

A1

3

- (b) (i) Char. eqn. is $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$
One A mark for each of the other coefficients

M1A1A1A1

ie $(\lambda - 2)(\lambda - 1)^2 = 0$
Good factorisation attempt

M1

giving $\lambda_1 = 1$ (twice) and $\lambda_2 = 2$

A1

6

- (ii) $\lambda = 1 \Rightarrow x - y + z = 0$ (thrice)

B1

Any two independent eigenvectors
Attempted

M1

(eg) $\alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

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$\lambda = 2 \Rightarrow -y + z = 0$
 $x - 2y + z = 0 \Rightarrow x = y = z$
 $x - y = 0$

M1

$y \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

A1

5

- (iii) For $\lambda = 1$, eigenvectors represent a plane of invariant points
Plane

M1A1

For $\lambda = 2$, eigenvectors represent an invariant line

B1
3
[20]

Q25.

(a) $\mathbf{AB} = \begin{bmatrix} 0 & -4 \\ 4 & 0 \end{bmatrix}$
M1A0 if 3 entries correct

M1A1
2

(b) $\mathbf{A}^2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

$\dots = 4\mathbf{I}$

B1
B1
2

(c) $(\mathbf{AB})^2 = -16\mathbf{I}$

B1

$\mathbf{B}^2 = 4\mathbf{I}$
PI

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B1

so $\mathbf{A}^2 \mathbf{B}^2 = 16\mathbf{I}$ (hence result)
Condone absence of conclusion

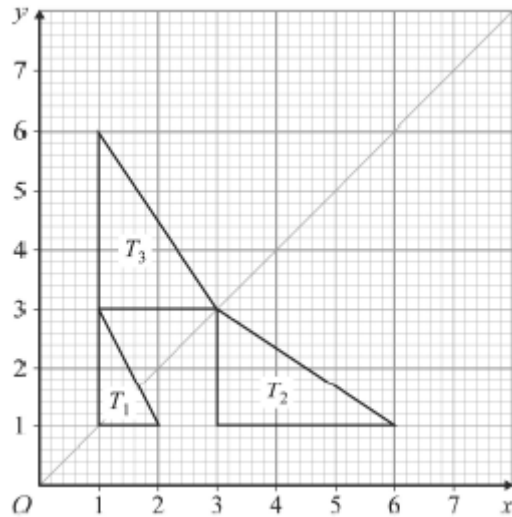
B1
3
[7]

Q26.

(a) Matrix is $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$
M1 if zeros in correct positions; allow NMS

M1A1
2

(b)



Third triangle shown correctly

M1A0 if one point wrong

M1A1

2

- (c) Matrix of reflection is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Alt: calculating matrix from the coordinates: M1 A2,1

B1

Multiplication of above matrices
in correct order

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Answer is $\begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$

ft wrong answer to (a); NMS 1/3

A1F

3

[7]

Q27.

- (a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} \sqrt{3} & 0 \\ 1 & 0 \end{bmatrix}$

M1A0 if 3 entries correct;

Condone $\frac{2\sqrt{3}}{2}$ for $\sqrt{3}$

M1A1

2

(ii) $\mathbf{BA} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

*Deduct one for each error;
SC B2,1 for AB*

B3,2,1

3

- (b) (i) Rotation 30° anticlockwise (abt O)
M1 for rotation

M1A1

2

- (ii) Reflection in $y = (\tan 15^\circ)x$
M1 for reflection

M1A1

2

- (iii) Reflection in x -axis
*1/2 for reflection in y -axis
ft (M1A1) only for the SC*

B2F

2

Alt: Answer to (i) followed by answer to (ii)
*M1A0 if in wrong order
or if order not made clear*

M1A1F

(2)

[11]

Q28.

- (a) A is a Rotation thro' 90°

M1 A1

about Ox

A1

B is a Reflection in $y = 0$ (i.e. $x - z$ plane)

M1 A1

5

(b) (i) $\mathbf{M}_C = \mathbf{M}_B \mathbf{M}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

M1 A1
2

- (ii) C is a Reflection in $y = z$
Give M1 for any series of reflections

M1

N.B. In (i): $\mathbf{M}_A \mathbf{M}_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ scores M0
but ft "Reflection in $y = -z$ " in (ii)

A1
2

[9]

Q29.

(a) (i) $\det \mathbf{P} = 4a + 6 + 4 + a = 5a + 10$

M1A1
2

(ii) When $a = 3$, $\det \mathbf{P} = 25$
ft

B1F
1

(iii) Setting their $\det \mathbf{P} = 0 \Rightarrow a = -2$

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ft

M1

A1F
2

(b) (i) $\mathbf{P}^{-1} = \frac{1}{25} \mathbf{Q}$

B1
1

(ii) $(\mathbf{PQ})^{-1} = (25 \mathbf{I})^{-1} = \frac{1}{25} \mathbf{I}$

M1A1
2

Or $(\mathbf{PQ})^{-1} = \mathbf{Q}^{-1} \mathbf{P}^{-1}$

Ignore $(\mathbf{PQ})^{-1} = \mathbf{P}^{-1} \mathbf{Q}^{-1}$ if they can make it work

(M1)

$$= \mathbf{Q}^{-1} \cdot \frac{1}{25} \mathbf{Q} = \frac{1}{25} \mathbf{I}$$

(A1)

(2)

(iii) $\det \mathbf{PQ} = \det (25 \mathbf{I}) = 25^3$ or 15625

M1A1

$\det \mathbf{PQ} = \det \mathbf{P} \cdot \det \mathbf{Q}$
Used

M1

$$\Rightarrow 25^3 = 25 \det \mathbf{Q}$$

$$\Rightarrow \det \mathbf{Q} = 25^2 \text{ or } 625$$

A1

4

[12]

Q30.

(a) $\mathbf{M} = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$

B1 if subtracted the wrong way round

B2,1

2

(b) $p = 3$

ft after B1 in (a)

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B1F

L is $y = -x$

Allow $p = -3$, L is $y = x$

B1

2

(c) $\mathbf{M}^2 = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$

Or by geometrical reasoning; ft as before

B1F

$\dots = 9\mathbf{I}$

ft as before

B1F

2

[6]

Q31.

- (a) (i) **AB** = a 3×3 matrix

M1

$$= \begin{pmatrix} 3 & 2 & t+1 \\ 1 & 2 & t-1 \\ 3 & 2 & t+1 \end{pmatrix}$$

At least 5 elements correct, incl. at least one from C_3

A1

All elements correct

A1

3

- (ii) **BA** = a 2×2 matrix

M1

$$= \begin{pmatrix} 2 & 2 \\ t & t+4 \end{pmatrix}$$

A1

2

- (b) $R_1 = R_2 (\Rightarrow \det \mathbf{AB} = 0)$

*Or expanding and **showing** $\det = 0$*

B1

1

- (c) **BA** = $\begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix} = 2\sqrt{2} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

M1 A1

E: enlargement s.f. $2\sqrt{2}$

B1

F: Rotation

NB: Rotation bit may be sorted completely separately in which case marks are split 3 + 3

M1

clockwise (about O) thro' 45°

Or -45° , 315°

A1 A1

6

[12]

Q32.

- (a) (i) Reflection ...

M1

... in $y = -x$
OE

A1

2

(ii) $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

M1A0 for three correct entries

M1A1

2

- (iii) $\mathbf{A}^2 = \mathbf{I}$ or geometrical reasoning

E1

1

(b) (i) $\mathbf{B}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

M1A0 for three correct entries

M1A1

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$\mathbf{b}^2 - \mathbf{A}^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$

ft errors, dependent on both M marks

A1ft

3

(ii) $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A}) = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$

B1

... = $\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

ft one error; M1A0 for three correct (ft) entries

M1

A1✓

3

[11]

Q33.

- (a) Attempt at either determinant

M1

$$\det \mathbf{P} = k - 2 \quad \det \mathbf{Q} = 3k - 28$$

A1A1

3

- (b) Use of $\det(\mathbf{PQ}) = (\det \mathbf{P})(\det \mathbf{Q})$

M1

Creating a quadratic

$$\text{From } (k - 2)(3k - 28) = 16$$

dM1

$$3k^2 - 34k + 40 = 0$$

A1✓

$$k = \frac{4}{3} \text{ or } 10$$

CAO

B1

4

Alternative:

$$\mathbf{PQ} = \begin{bmatrix} 28 & 2k+15 & 3 \\ 7k+2 & 2k+4 & 2k+2 \\ 48 & 3k+26 & 6 \end{bmatrix}$$

B1

$$\det(\mathbf{PQ}) = 3k^2 - 34k + 56$$

B1✓

Equating to 16 and solving

M1

$$k = \frac{4}{3} \text{ or } 10$$

CAO

A1

4

[7]

Q34.

(a) (i) $M^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

M1 if 2 entries correct

M1A1 if 3 entries correct

M1

A2,1

3

(ii) $M^4 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

ft error in M^2 provided no surds in M^2

B1ft

1

(b) Rotation (about the origin)

M1

... through 45° clockwise

A1

2

(c) Awareness of $M^8 = I$

OE; NMS 2/3

M1

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$M^{2006} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

complete valid method

m1

ft error in M^2 as above

A1ft

3

[9]

Q35.

(a) Setting $\det \mathbf{P} (2t - 6 + 2 - 3t + 8 - 1 = 3 - t) = 0$

M1

$\Rightarrow t = 3$

A1

2

(b) (i)
$$\begin{bmatrix} 6 & 0 & 0 \\ -7t-21 & 3-t & 15+5t \\ 0 & 0 & 6 \end{bmatrix}$$

Multiplication attempt with ≥ 3 correct elements

M1

Any one row or column

A1

All correct

A1

3

(ii) When $t = -3$, $PQ = 6I$

The 6 may be implicit

B1 B1

2

(iii)
$$Q^{-1} = \frac{1}{6}P = \frac{1}{6} \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & -2 \\ 3 & 2 & 1 \end{bmatrix}$$

Allow for $\frac{1}{6}P$ or $\frac{1}{36}(6P)$

B1

1

(c) Enlargement (centre O) sf 6

M1 A1

2

[10]

Q36.

(a) Transformation is a reflection in $y = x$

B2

2

(b) Matrix is
$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

M1 for matrix for a rotation;

M1

A1 for correct trig expressions

A2,1

3

- (c) Attempt to multiply the matrices

M1

... in the correct order

m1

Matrix is $\begin{bmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

Wrong answer to (b)

A1ft

3

[8]

Q37.

(a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 3 & 6 \\ 6 & 3 \end{bmatrix}$

M1A0 if 3 entries correct

M1A1

2

(ii) $\mathbf{AB} = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$

Ditto

M1A1

2

(b) $\mathbf{A} + \mathbf{B} - \mathbf{AB} = \begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}$

ft wrong answers in (a)

B1F

... = $-5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B1

2

[6]

Q38.

(a) **A:** Reflection in $y = z$

M1 A1

B: Reflection in $y = 0$ (x - z plane)

M1 A1

4

(b) (i) $\mathbf{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$
 ≥ 5 entries correct

All correct

B1

B1

2

(ii) About the x -axis; through 90°
 $+/-$; or 270° ; or in radians

B1B1

2

[8]

Examiner reports

Q1.

The vast majority of students selected the correct answer of **AC**. The most common incorrect choice was **AB**.

Q2.

Only half of the students were able to make satisfactory progress with this question. Most

of the successful attempts substituted $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (or equivalent) for **A**, and then solved the resulting simultaneous equations. Only the most able students were able to successfully isolate **A** or **A**⁻¹ by rearranging the given matrix equation.

Q4.

In this question on matrices, parts (a)(i) and (a)(ii) were almost always answered correctly,

although some students left their answer to (a)(ii) in the form $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ without indicating that they clearly appreciated what **I** represented. A correct description of the geometrical transformation represented by **A** was usually given although reflection in the line $y = x$ was the most frequent incorrect answer.

Most students scored the method mark in part (b)(ii) but some then gave the incorrect value $\sqrt{2}$ for k . Part (b)(iii) proved to be more of a challenge. A significant minority of students could not make any real progress. Most realised that the matrices **A** and **B**

needed to be multiplied but the product was not always linked with the matrix for **P**, $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$, in a correct manner. Some better approaches just made the mistake of using **BA** instead of **AB**. For the final mark the examiners wanted the coordinates of the image of **P** to be written in coordinate form rather than left as a matrix.

Q5.

This question elicited a good response from students. There was evidence of efficient use of graphic calculators in parts (a) and (c). In part (b), students had to produce a watertight proof to get both marks. It was disconcerting to find students writing “divide by **A**” or “cancel **A**”, displaying a lack of understanding matrix algebra. In the final part, many chose not to use the inverse matrix to find the solutions and instead chose to use elimination. Varying degrees of success were seen – the most successful chose to eliminate y from the second and third equations.

Q6.

Many correct answers were submitted for part (a) in this question which tested 2×2 matrices including the identity matrix **I**. In part (b) the most successful starting approach was to find the matrix for **A – B + AB** in terms of p , equate it to the matrix for $k\mathbf{I}$ to show that $p = -7$ and then to work with **A – B + AB** and show that it was equal to $-27\mathbf{I}$. A minority of students found the correct value of k but either did not convincingly show that **A – B + AB** = $k\mathbf{I}$ or made an incorrect concluding statement. It was disappointing to see some students writing down an incorrect matrix for **I**.

Q7.

A good response was received to this question. In part (a) it is not sufficient to give the determinant as 3 and deduce that the matrix is non-singular. The key point to secure the mark is to state that $3 \neq 0$ and then make the deduction. Inverse matrices are well understood, marks were dropped here for the standard types of errors with signs or miscalculation of minor determinants. Part (c) proved more differentiating. A minority of students were not aware of the result $(AB)^{-1} = B^{-1}A^{-1}$ and hence set off on a trail to find B^{-1} by other, often invalid, means.

Q8.

Parts (a)(i) and (a)(ii) were generally answered correctly although some rectangles for R_3 were drawn which had vertices with either the wrong x -coordinates or the wrong y -coordinates.

The most common wrong answer for part (b)(i) was the matrix which represented a rotation of 90° anticlockwise and in part (b)(ii) it was not uncommon to see the matrices being multiplied in the wrong order. A small number of candidates gave solutions which had matrices left in terms of trigonometric functions; these were penalised.

Q9.

This was the least well-answered question on the paper with only the most able scoring full marks. Part (a) was generally well-answered with only a small minority failing to give the matrix in surd form. In part (b)(i), better students showed clear working which related to part (a), and stated correct answers for both the scale factor of the enlargement and the angle of the rotation.

In part (b)(ii) the most common method seen was to find the matrix M^2 first and use it to find the scale factor and angle rather than stating the scale factor to be the square of the scale factor in part (b)(i) and the angle to be double the angle in part (b)(i). Those using the matrix M^2 usually gave the correct scale factor but 90° anticlockwise was a common wrong angle.

Students generally scored both marks in part (b)(iii), although some others made an arithmetic error in their matrix multiplication or did not complete the question appropriately and thus lost the final mark. Part (b)(iv) proved to be very challenging and only a small number showed the deduction sufficiently rigorously. There were many examples of students obtaining the correct value of n but from incorrect reasoning. Most attempted this part using algebraic methods but relatively few gave due consideration to the negative sign.

Some excellent deductions based on geometric reasoning using the earlier results in the question were also seen.

Q10.

Apart from the few who did not seem to know what a transpose matrix was, this was a popular and high-scoring question, even for those with relatively low scores on the paper as a whole. The only really disturbing aspect of the work on display was that in part (b)(ii) so few students seemed to have any idea how to go about finding a linear factor of a fairly simple cubic and then factorising to find the accompanying quadratic factor, which is C1 work. Most students who gained the answer $k = -3$ appeared to get it from a calculator, and very few indeed could use it to find the required quadratic factor; fewer still were able to reason convincingly that this quadratic factor had no real roots, which again is C1 work.

In the final part of the question, many students failed to realise that k was now being replaced in the given matrix $\mathbf{A}$ by $k - 7$, so that all they had to do was to add 7 to their previous value of k for two straightforward marks. Almost all students who correctly found the answer for this new k did so by going back, working out a cubic characteristic equation (with some large numbers for coefficients), and using a calculator to solve it for them. This was, somewhat reluctantly, awarded the marks if the answer was correct. Most of those students who did not get the correct answer also used this method, but any incorrect working prevented their calculator helping them out and so they gained no marks.

Q11.

Like the preceding questions, this one produced a good opportunity for most candidates to score high marks. The first two marks were sometimes lost because the candidate failed to provide numerical values for the sines and cosines in their matrices. Also the first matrix was often that of a 90° anticlockwise rotation, rather than a clockwise one as required. Most candidates earned two marks in part (b)(i), relatively few finding $\mathbf{BA}$ instead of $\mathbf{AB}$. Full marks were very common in part (b)(ii). The geometrical interpretations asked for in part (c) were mostly correct, though some floundered somewhat in the first part. In part (c)(ii), the answer 'enlargement with scale-factor -25 ' was acceptable, and very common.

Q12.

This was another popular question, in that responses were generally at least partially successful. Even so, around half of all candidates failed to realise that the "hence" in part (a)(ii) meant that they were supposed to work algebraically with the result of part (a)(i) rather than go through the lengthy approach of finding $\mathbf{X}^{-1}$ directly. About the same proportion incorrectly opted for $(\mathbf{XY})^{-1} = \mathbf{X}^{-1}\mathbf{Y}^{-1}$. In addition to all these errors, a lot of marks were lost due to carelessness; mistakes made in calculating $\mathbf{X}^2$ were often not corrected, even when the candidates clearly failed to arrive at a multiple of $\mathbf{I}$ in part (a)(i).

Then, despite being told that $\mathbf{X}^{-1}$ could be found by $\frac{1}{20}(\mathbf{X} - \mathbf{I})$ in part (b), many made a slip

somewhere in the working, and the correct $(\mathbf{XY})^{-1}$ appeared far less often than should have been the case, given the information given in the question.

Q13.

The first matrix calculation in this question involved a fair number of surds and minus signs, and yet it was very accurately performed by the majority of candidates. After that, it was relatively easy to reach a matrix of the required form in part (a)(ii). Almost all candidates knew how to describe the transformation represented by this matrix, but by contrast many had a long and unsuccessful struggle to describe the transformation required in part (b)(ii). An acceptable alternative here was an enlargement with scale factor -2 and a rotation through 60° clockwise.

Q14.

As usual on this paper, the work on matrices was very good indeed, with most candidates working out all the steps efficiently. Some tried to expand the expressions $(\mathbf{A} - \mathbf{I})^2$ and $(\mathbf{A} - \mathbf{B})^2$ but almost invariably assumed commutativity of multiplication. A rather silly way to lose a mark was to work correctly to the equation $3 - p = 12$ and then to solve this equation incorrectly, which happened quite frequently.

Q15.

The matrix multiplication was generally handled very well, apart from a small but noticeable number of candidates who insisted on attempting **BA** despite the question. The only other obstacle to a completely correct part (a) was found amongst those candidates who had the odd one or two incorrect elements, which they had failed to notice and correct due to a lack of a check, meaning that $t = 7$ did not actually give **4I** consistently for their **AB**. As with question 2, the word “deduce” in part (b) was almost totally ignored, and alternative methods for finding an inverse, often taking up lots of time, received no credit.

Q16.

Rather bizarrely, the final bit of part (a) proved to be the most common difficulty on the paper. Having found the image of $(x, 2x + 1)$ to be $(x - 1, 2 - x)$, hardly anyone managed to deduce that the equation of the image-line was $y = 1 - x$. Parts (b) and (c) were handled very well indeed, although a few candidates tried **BA** rather than **AB** in part (c).

The message as to what is an acceptable way to describe a shear has been stated often in recent years’ reports, and this message has clearly been picked up by nearly all centres: the mapping of a point (not on the line of invariant points) to its image was noted far more regularly than in previous sessions, though there are still some rather spurious references to scale factor — ‘spurious’ in that it is not easy to assign to it an obvious significance or role in the proceedings; indeed, it is not easy even to decide what it is in cases where the shear is not parallel to one of the coordinate axes.

Q17.

High marks were often earned in this question, generally from the multiplication of the matrices rather than from the geometrical explanations, which tended to be shaky. In parts (c) and (d) the vast majority of candidates calculated a matrix product rather than base their answers purely on the transformations already found in parts (a) and (b). The transformation in part (c) was often given as a reflection rather than a rotation, and in part (d) many candidates stated that the matrix was the identity matrix but did not make any geometrical statement as to what this matrix represented. In part (e) the correct matrix was often obtained but the candidates failed to give the correct geometrical interpretation, or in some cases resorted to a full description of the transformation as a combination of the reflection and rotation found in parts (b) and (a). When this was done correctly, full credit was given.

Q18.

This was another straightforward question and was usually well-received and successfully attempted. In part (b), candidates seemed evenly split between those who recognised that $\mathbf{B}^T \mathbf{A}^T = (\mathbf{AB})^T$ and those who simply worked out $\mathbf{B}^T \mathbf{A}^T$ directly. However, the very last mark

on the question was often not gained, as many failed to show *visibly* that $x = \frac{1}{2}$ worked in **both** of the equations that arose when comparing elements of the two matrices.

Q19.

The first nine marks in this question seemed to be found very easy for most candidates to earn. The presence of so many k ’s in the matrices did not prevent them from carrying out all the calculations correctly and confidently, though sometimes the flow of the reasoning in part (b) was not made totally clear. Part (c)(ii), on the other hand, proved too hard for many candidates, who seemed to resort to guesswork for their answers.

Q20.

This question was intended to test a couple of basic areas of the module's work on determinants and transposes. The former topic was generally handled very easily, but a majority of candidates appeared to resort to guessing what was involved in part (b) with the work on transposes. In general, candidates scored either 3 or 6 marks on this question.

Q21.

In this question part (a) was very well answered in general, candidates being able to choose the correct form of matrix for each type of transformation and to insert the correct forms for the necessary sines and cosines.

In part (b), however, the majority of candidates failed to see the close connection between the given matrix and the one that they had found in part (a) (ii). The answers given often appeared to be wild guesses.

In part (c) many candidates saw the need to multiply the two matrices but often in the

wrong order. Those who obtained the correct matrix $\begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$ when giving the geometrical interpretation.

Q22.

This was a straightforward starter to the paper, and was generally found to be so by candidates.

Almost all candidates knew what to do, in principle, although there were unexpectedly large numbers of arithmetical (particularly sign) errors in part (b) when finding the required value of k .

Q23.

Part (a)(i) of this question was almost universally well answered, but part (a)(ii) led to some very slipshod and unclear reasoning. In part (b)(i) many candidates did not appreciate that the scale factor of the enlargement must be the same as the value of q obtained in the previous part. In part (b)(ii) a common mistake was to use $\tan 60^\circ$ in finding the gradient of the mirror line instead of halving the angle to obtain $\tan 30^\circ$. Answers to part (c) were mostly very good, even from candidates who had been struggling with the earlier parts of the question.

Q24.

This question was definitely the most demanding one on the paper; partly for its length, and partly due to working with a 3×3 matrix. On top of this, there was a repeated eigenvalue to the matrix, and a lot of candidates clearly did not know quite what to make of it. Nonetheless, even the relatively weaker candidates were still able to gain up to 12 of the marks.

Part (a)(i) was very well handled by those who attempted it. Part (a)(ii) was less well handled. Many of those who fell down here did so because they then went off to find the inverse matrix $\mathbf{M}^{-1}$ itself, rather than follow the logic of the question. Many more who did multiply the matrix equation by $\mathbf{M}^{-1}$ got tangled up in some way – failing to spot that $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$, for instance, or being unable to divide by 2 under exam conditions.

In part (b), the factorisation of the determinant $|\mathbf{M} - \lambda\mathbf{I}|$ was usually approached by

row/column operations, which was very pleasing to see. However, it frequently (actually usually) led to the strange sight of candidates taking a partially or completely factorised expression, and then multiplying it out to get a standard cubic form, only to start to factorise it again to find the three eigenvalues.

Those who made a mistake in this working then shot themselves in the foot, losing the following method mark for the attempt to factorise, which they failed to pick up by presuming that their cubic actually was equal to the factorised form $(\lambda - 1)^2(\lambda - 2)$ rather than doing any of the necessary working or going back to check their previous, incorrect working.

The final two parts to the question were demanding, but actually not as difficult as candidates found them to be. Despite the fact that most candidates getting this far quite correctly deduced that $x - y + z = 0$ for the repeated eigenvalue $\lambda = 1$, almost all of them then believed this represented the equation of a line. This error then prevented them from finding a pair of independent representative eigenvectors (or an alternative parametric plane form) and also from describing the result as a plane of invariant points, rather than as a line.

Q25.

This question was very well answered by the majority of candidates, who showed confidence and accuracy in manipulating these simple matrices.

Q26.

This question was not generally well answered, except for part (b) where most candidates were able to draw the reflected triangle correctly on the insert. In part (a) many candidates seemed not to recognise the transformation as a simple one-way stretch, and even those who did were not always familiar with the corresponding matrix. They could have worked out the matrix by considering the effect of the stretch on the points (1, 0) and (0, 1), but in many cases candidates either gave up, made a wild guess, or embarked on a lengthy algebraic process involving four equations and four unknowns.

In part (c) relatively few candidates saw the benefit of matrix multiplication for the composition of two transformations, and even they often multiplied the matrices the wrong way round. Again it was common to see candidates trying to find the matrix from four equations, but this method was doomed to failure if the candidate, as happened almost every time, paired off the vertices incorrectly.

Q27.

This question was very productive for the majority of candidates, who showed a good grasp of matrices and transformations. A strange error in part (a)(i) was a failure to

$$\frac{2\sqrt{3}}{2}$$

simplify the expression $\frac{2\sqrt{3}}{2}$, though on this occasion the error was condoned. The most common mistake in part (a)(ii) was to multiply the two matrices the wrong way round. Only one mark was lost by this as long as the candidate made no other errors and was able to interpret the resulting product matrix as a transformation in part (b)(iii). Occasionally a candidate misinterpreted the 2θ occurring in the formula booklet for reflections, and gave the mirror line as $y = x \tan 60^\circ$ instead of $y = x \tan 15^\circ$.

Q28.

This question relied on the use of information given in the formulae booklet, and the

majority of candidates coped very well with it. Many fell at the obvious hurdle in part (b), by supposing that “A followed by B” was represented by the matrix $\mathbf{M}_A\mathbf{M}_B$, rather than $\mathbf{M}_B\mathbf{M}_A$. Candidates who failed to attempt to describe the three transformations by a single, rather than multiple (ie composite), description were not awarded marks. There was no penalty for candidates who, for instance, called a plane a line, so long as they gave the right equation.

Q29.

This was the last question on the paper because, since it involves a 3×3 matrix, it was potentially the most time-consuming question. However, candidates found the structure of the question very helpful.

Q30.

This question was generally well answered. The most common errors were in part (b), where the mirror line was given as $y = x$ rather than $y = -x$. It was, however, acceptable to give this reflection in conjunction with an enlargement with scale factor -3 .

In part (c) some candidates omitted the factor 3 before carrying out the matrix multiplication, while others multiplied incorrectly to obtain the 9s and 0s in all the wrong places.

Q31.

The two product matrices $\mathbf{AB}$ and $\mathbf{BA}$ in part (a) of this question were very popular and the majority of candidates gained all 5 marks. The rest of the question really was poorly done on the whole, but this was the most demanding work on the paper. In part (b), many of candidates noted that $\det(\mathbf{AB})$ was zero, but failed to explain why. In part (c), most seemed to resort to guesswork, especially regarding the transformation F. A shear was, marginally, the most popular choice, with a 90° rotation close behind. The majority of candidates seemed to have no method of approach to the problem. Those that took a scalar factor out of the matrix usually contented themselves with 2, 4 or 8 as the scale factor of the enlargement E; thereby leaving themselves with a matrix they simply did not know what to do with. Those who chose “rotation” as their answer then did so from a matrix with $\det \neq 1$ and consequently did not gain all the marks available.

Q32.

Many candidates were successful in part (a)(i) of this question. Wrong answers usually involved a 180° rotation or a reflection in $y = x$. Part (a)(ii) was nearly always done correctly, but the explanations offered in part (a)(iii) were often wrong or too vague to be worth a mark. The matrix calculations in part (b) were done well apart from occasional careless slips. Some candidates seemed to think that three marks were being offered in part (b)(ii) for merely quoting the same answer as in part (b)(i), while others attempted to use matrix algebra to express $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A})$ in an alternative form before evaluating their expression. This usually led to a wrong answer.

Q33.

This was another straightforward question, and almost all candidates scored highly on it. Apart from a very few who multiplied to get $\mathbf{PQ}$ first, most were happy to use the result $\det \mathbf{PQ} = (\det \mathbf{P})(\det \mathbf{Q})$ to get the required quadratic and find the two values of k .

Q34.

Most candidates carried out the matrix multiplications accurately in part (a) of this question. In part (b) many candidates stated correctly that the required transformation was a rotation, but often gave the angle as 45° anticlockwise rather than clockwise. Attempts at part (c) were rarely completely successful. Many candidates found that $\mathbf{M}^8$ was the identity matrix but then went on to say that $\mathbf{M}^{2006}$ must also be the identity matrix. Others gave the same answer as in part (a) (i), or gave the correct answer without any indication as to their reason for doing so, or with incomplete reasoning.

Q35.

This proved to be a profitable question for many candidates. In particular, the multiplication of two 3×3 matrices was handled very competently indeed, even though algebraic terms were involved. The odd error arising here did not prevent candidates from doing good work in the later parts of the question. They were not required to check that all entries of the product matrix gave consistent values for both t and k .

Q36.

The responses to parts (a) and (b) of this question were most impressive, showing that a good majority of candidates were familiar with matrices of simple transformations. Unfortunately, there were relatively few candidates who knew which way round to multiply the matrices in part (c). Even very strong candidates often went wrong here.

Q37.

Most candidates made a good start to the paper by answering this question confidently, although in part (c) many candidates failed to show knowledge of the identity matrix.

Q38.

Many Candidates would have benefited greatly from a keener awareness of what is given them in the **Formulae Booklet**. This was really a very direct test of those ideas.

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